SUPPORTING ADULT GUIDANCE

Repeated change: compound interest and depreciation

Percentages, Ratio, Growth and Bounds · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The lesson opens with two offers that add the same £50 in year 1 but end differently after 10 years. The main activity, the Year-by-Year Race, is in Movement. Learners choose savings (interest) or a car (depreciation), set the starting amount and rate, and step through the years. Each year draws a pair of columns: the same amount each year (simple interest or straight-line loss) and the same percentage each year (compound interest or depreciation), with the gold difference on top. The readout shows each year's multiplication, how much was earned or lost and why it changes, then the power form (for example 800×1.03^4). A table lists every year. Money is shown to the nearest penny and labelled as such; where the exact value is short it is shown too (for example £562.43 to the nearest penny, exactly £562.432). Six questions, a stretch (where the extra £4 comes from), true-or-not statements and planning briefs follow.

Driving Question: *What if the change happens again and again?*

The mathematics, in plain terms

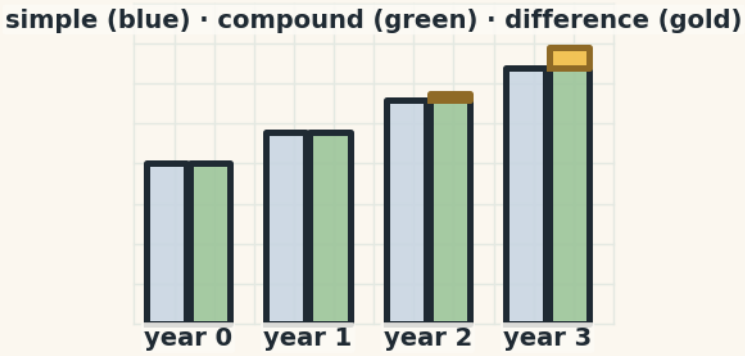
Compound growth: $A = P(1 + r)^n$; simple interest: $A = P(1 + nr)$. For $r > 0$ and $n \geq 2$, $(1 + r)^n > 1 + nr$ (Bernoulli's inequality), so compound interest is ahead from year 2. The first extra term, $P \times n(n - 1)/2 \times r^2$, is the interest on interest. Depreciation uses $(1 - r)^n$, which is never 0 for $r < 1$.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Compound interest

£400 is saved at 20% compound interest for 3 years. How much is there? How does it compare with simple interest?

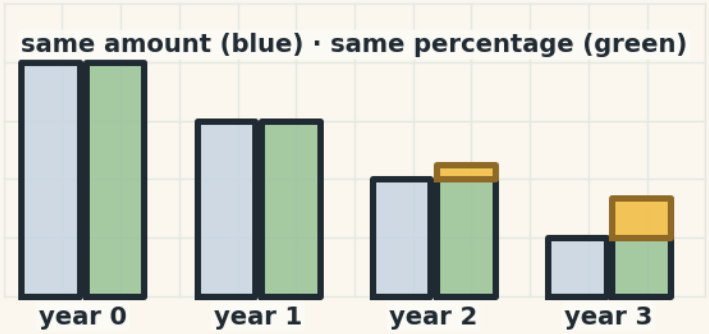


1. The multiplier for one year	100% + 20% = 120% = 1.2
2. Year by year	400 × 1.2 = 480; 480 × 1.2 = 576; 576 × 1.2 = 691.20
3. As a power	$400 \times 1.2^3 = 400 \times 1.728 = 691.20$
4. Simple interest	400 + 3 × 80 = 640

£691.20 with compound interest: £51.20 more than simple interest, because of interest on interest.

Example 2 — Depreciation

A bike worth £800 depreciates by 25% a year. What is it worth after 3 years?



1. The multiplier for one year	100% – 25% = 75% = 0.75
2. Year by year	800 × 0.75 = 600; 600 × 0.75 = 450; 450 × 0.75 = 337.50
3. As a power	$800 \times 0.75^3 = 800 \times 0.421875 = 337.50$

£337.50. (Losing £200, 25% of the original, every year would leave only £200.)

Step the years, then see the power

Learners first multiply one year at a time, so the power is a short way to write something they have already done. The two columns side by side make the difference between adding the same amount and multiplying by the same amount visible before it is named.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Which would you choose? Two offers; the idea card.	~5 min
Movement	The Year-by-Year Race (main activity), six questions, the stretch.	~20 min
Reflection	True or not? Repeated percentages multiply.	~6 min
Creativity	Plan it: six briefs.	~9 min
Rest	Small steps that build on each other.	~2 min
Nutrition	Practical: reading savings offers and values that fall.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding the percentages: 3 years at 4% is 12%.
Ask: “Is the 4% in year 2 a percentage of the same amount as in year 1?”

Using simple interest when the question says compound.
Ask: “Does the interest earn interest?”

Multiplying by the number of years instead of using a power (500 × 1.04 × 3).
Ask: “How many times do you multiply by 1.04?”

Using 1.15 for a 15% decrease.
Ask: “What percentage of the value is left?”

Rounding each year and carrying the rounding on.
Ask: “Can you keep the full value until the end?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: percentages as multipliers (Lesson 3) and powers. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says £1000 at 10% compound interest for 3 years gives £1300. The aim is a question that brings out interest on interest.

- “How much is there after 1 year?”
- “In year 2, 10% of what?”

- “So how much interest in year 2? Is it the same as year 1?”
- “How could you write three years of × 1.1 in one go?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does the table say “to the nearest penny”?

Repeated multiplication quickly gives values with many decimal places. The lesson rounds money to the nearest penny and says so; where the exact value is short, the readout shows it too.

Should learners round each year?

Encourage keeping the full value and rounding once at the end. Answers to the nearest penny are accepted.

Is the stretch required?

No. The secure core is multiplying by the multiplier once per year, or using a power.

What comes next?

Lesson 7: growth models, comparing adding the same amount (linear) with multiplying by the same amount (exponential).

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.